

12-Week Practice Plan — Class 4 Mathematics Olympiad

Built around the SOF IMO pattern (35 Q / 40 marks / 60 min — see [02-exam-pattern.md](#)). Three phases: build core skills topic by topic, add pure logical reasoning, then stress-test with full timed papers.

Suggested pace: **3 short sessions a week, ~30 minutes each**, not one long cram — olympiad questions reward familiarity and speed more than raw study time.

Phase 1 — Core Skills (Weeks 1–8)

One topic per week. Each week = 1 assignment of ~15 MCQs covering that topic's direct-computation + word-problem flavours (i.e. Mathematical Reasoning + Everyday Mathematics style questions together), plus a short self-check against the answer key.

Week	Topic	Folder
1	Number Sense & Numerals (place value, expanded form, ordering, rounding, Roman numerals)	assignments/01-number-sense/ ✓ built
2	Four Operations (+ − × ÷, multi-digit, word problems)	assignments/02-four-operations/
3	Factors, Multiples & Number Patterns	assignments/03-factors-multiples-patterns/
4	Fractions & Decimals (comparing, equivalent, simple operations)	assignments/04-fractions-decimals/
5	Measurement: Length, Weight, Capacity + Money	assignments/05-measurement-money/
6	Time & Calendar	assignments/06-time-calendar/
7	Geometry (shapes, lines, symmetry) + Perimeter & Area	assignments/07-geometry-perimeter-area/
8	Data Handling (tally marks, pictographs, bar graphs)	assignments/08-data-handling/

Weekly routine: attempt the assignment untimed first; mark it against the answer key; redo only the missed questions 2 days later to confirm the concept actually stuck.

Phase 2 — Logical Reasoning (Weeks 9–10)

Pure Section-1 skills, since these don't show up anywhere else in the school curriculum and need dedicated practice.

Week	Topic	Folder
9	Patterns, Analogy & Classification, Alphabet Test, Coding–Decoding	<code>assignments/09-reasoning-patterns-coding/</code>
10	Ranking Test, Direction Sense, Days & Dates, Mirror Images, Embedded Figures	<code>assignments/10-reasoning-direction-figures/</code>

Note on figure-based items (mirror images, embedded figures, geometrical solids): these need actual diagrams. The text-based assignments here substitute equivalent letter/number/logic puzzles for the figure-reasoning skills that *can* be done in text (series, coding, ranking, direction, calendar), and flag the diagram-only question types separately so they can be sourced from an image-capable worksheet generator or workbook — see the note at the bottom of `assignments/README.md`.

Phase 3 — Achievers / HOTS + Full Papers (Weeks 11–12)

Week	Activity
11	<code>assignments/11-achievers-hots/</code> — 2-and-3-step mixed problems pulling from every Phase-1 topic. Then attempt Sample Paper 1 (untimed, open-book) to find the gaps — see <code>sample-papers/README.md</code> .
12	Attempt Sample Papers 2–6 , each strictly timed to 60 minutes , one every 1–2 days. After each paper, log every wrong answer in an error log (topic + why it was wrong) before moving to the next paper.

Sample paper schedule (Phase 3)

Paper	When	Mode
Sample Paper 1	Week 11, end	Untimed, review-focused
Sample Paper 2	Week 12, Day 1	Timed (60 min)
Sample Paper 3	Week 12, Day 3	Timed (60 min)
Sample Paper 4	Week 12, Day 5	Timed (60 min)
Sample Paper 5	Week 12, Day 7	Timed (60 min) — treat as Level-1 dress rehearsal

Paper	When	Mode
Sample Paper 6	Week 12, Day 9	Timed (60 min), Level-2 difficulty step-up, for students likely to qualify

Only Sample Paper 1 is built out in this repo so far (full 35 questions, 40 marks, answer key) — see `sample-papers/README.md` for the plan to add 2–6 in the same format.

Progress tracker

Copy this checklist and tick off as you go:

- ☐ Week 1 — Number Sense
- ☐ Week 2 — Four Operations
- ☐ Week 3 — Factors, Multiples & Patterns
- ☐ Week 4 — Fractions & Decimals
- ☐ Week 5 — Measurement & Money
- ☐ Week 6 — Time & Calendar
- ☐ Week 7 — Geometry, Perimeter & Area
- ☐ Week 8 — Data Handling
- ☐ Week 9 — Reasoning: Patterns/Analogy/Coding
- ☐ Week 10 — Reasoning: Ranking/Direction/Calendar
- ☐ Week 11 — Achievers/HOTS + Sample Paper 1
- ☐ Week 12 — Sample Papers 2–6 (timed)

Optional extension

Once the 12 weeks and all 6 sample papers are comfortably done (consistently $\geq 32/40$, i.e. Level-2 cutoff range), move to Math Kangaroo-style puzzle questions for an extra spatial-reasoning/creative-problem-solving stretch — see the note in `01-syllabus.md` §3.